OMNICELLULAR PHYSIOLOGY

Full Independent Research Execution

Eight Peer Gen-1 Architecture Families

Document control	Value
Prepared for	Nathan Lewis, Project Architect
Prepared by	Microsoft 365 Copilot
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Status	Research portfolio for Architect review, not canon
Scope	Eight peer Gen-1 concepts; physical and canon tracks separated
Safety boundary	Conceptual and non-operational; no harmful biological or exposure procedures

Purpose, Origin, Mission, and Intended Accomplishment

Purpose

This document executes the complete Microsoft 365 Copilot v2.0 directive. It proposes eight independent Omnicellular Physiology architecture families, evaluates them under common evidence and threat standards, exposes their failure boundaries, and recommends a research portfolio rather than a consensus winner.

How the Topic Arose

The project began as a multi-scale simulation program in which 1AI builds models and -1AI challenges them. The supplied roadmap targets simulation of 2,000 bodies through abstraction rather than literal per-cell computation. The later physiology request moved from simulation of bodies to independent design of possible embodied architectures, with explicit concerns about hostile confinement, capture, unauthorized replication, nonstandard environments, longevity, and the ten-category extinction taxonomy.

Mission Statement

What This Research Aims to Accomplish

- Explore the architecture space before cross-AI convergence.
- Separate established mechanisms, engineering extrapolations, speculation, and canon-only proposals.
- Treat identity continuity and capture resistance as equal to mere biological survival.
- Produce simulation-ready state variables and falsification criteria.
- Identify failure modes early enough to guide a Gen-2 program.
- Provide a strong narrative body and a dedicated visual appendix.

Executive Summary

The research rejects the premise that “Omnicellular” should mean every cell performs every function simultaneously. That interpretation creates volume, reaction-compatibility, heat, signaling, and single-cell theft problems. The working definition used here is latent universal competence plus governed specialization: cells or modules can assume multiple roles, but capability is compartmentalized, permissioned, and frequently distributed across unlike units.

The eight families deliberately make different survival bets. Mosaic Covenant prioritizes heterogeneous cell guilds. Ceramic Lattice prioritizes mechanical continuity. Mycelial Relay prioritizes reroutable network topology. Phase-Core Mantle prioritizes thermal and chemical buffering. Archive Swarm prioritizes physical distribution. Electrogenic Reef prioritizes continuous environmental energy harvesting. Chrysalis Ledger prioritizes safe dormancy. Axiom Shell prioritizes honest separation between physical engineering and canon-only mechanisms.

No family satisfies every requirement without tradeoffs. The recommended physical-track portfolio is Mosaic Covenant as Primary, Chrysalis Ledger as Reserve, and Archive Swarm as Wildcard. Phase-Core Mantle should be retained as a subsystem donor for thermal design. Axiom Shell should remain the canon-governance wrapper rather than the baseline physical body.

Round 0: Prompt Audit and Amendment Log

Source Register

Scientific Baseline and Evidence Discipline

Synthetic-cell research has not produced a general-purpose synthetic organism of the type proposed here. Contemporary reviews emphasize that even defining a synthetic cell is unsettled, while common minimal ingredients include a selective boundary, metabolism supporting homeostasis, and an information carrier. The architectures below therefore use synthetic-cell ideas as scaffolding, not demonstration of feasibility.

Telomeres protect chromosome ends, telomerase can extend telomeric DNA, and telomere attrition is associated with senescence. Removing replication limits alone is not a safe longevity design because tissue control, mutation, metabolism, and cancer-like runaway remain distinct problems. Hypoxia-tolerant organisms often suppress metabolism and ATP demand rather than obtaining energy from nothing. Heat-shock proteins support proteostasis under stress, but do not eliminate the need for a complete heat balance. Tardigrade radiation tolerance and Dsup-related DNA protection provide useful biological motifs, while NASA continues to treat space radiation as a significant health risk requiring mitigation. Distributed multicellular computation has been demonstrated in limited synthetic-biology systems, but scaling it to identity-bearing whole-body cognition is speculative.

> ***Public research basis: synthetic cells [Nature Reviews Molecular Cell Biology, 2023]; telomeres [Cells, 2019]; hypometabolic hypoxia response [Metabolites, 2021]; heat-shock response [Int. J. Molecular Sciences, 2024]; tardigrade radiation tolerance and Dsup [Cancers, 2019; eLife, 2019]; space radiation [NASA]; distributed biological computation [Nucleic Acids Research, 2022]. Full URLs are listed in References.***

Operational Definition and Physiological Constitution

Three Laws Kept Across All Eight Families

1. Conservation Law: physical-track designs conserve mass-energy and must identify heat and waste destinations.
2. Causal Continuity Law: identity claims require an unbroken causal process or are labeled restoration, successor, or duplicate.
3. Authenticated Consent Law: no external link, sample, archive, or peer automatically gains authority to modify identity-bearing systems.

Shared Physical-Track Rules

- RESP: energy pathways must identify inputs, outputs, reserves, waste, debt, and failure state.
- RENEW: renewal must separate chromosome-end maintenance from mutation control, lineage authority, and runaway prevention.
- THERM: every operating state must close a first-law heat balance and identify the first saturating sink.
- CONV: information and templates do not substitute for feedstock; repair consumes matter and energy.
- NET: links are advisory and authenticated; disconnection cannot equal death.
- ID: repair and dormancy may preserve continuity; reconstruction and duplication require separate classification.

Eight Independent Architecture Dossiers

Architecture A: Mosaic Covenant

Core wager: Survival improves when no single cellular exploit can generalize across the whole body. Principal sacrifice: greater coordination overhead and slower global reconfiguration.

Architecture and Omni-cell Model

Substrate: biohybrid cell guilds sharing a common interface but using different internal chemistries. Matrix: interlocking tissue mosaics with rotating guild membership. The design treats omniscience at the cell level as a liability. Universal competence is latent and constrained by local role, environmental state, authorization, and resource availability.

Hereditary, Energy, Thermal, Repair, Identity, and Link Systems

Three Observation Rules

- OBS-A1 Triangulated Reality Rule: irreversible conclusions require dissimilar sensor classes.
- OBS-A2 Self-Model Dissent Rule: local cells may challenge the body-state estimate without overriding it.
- OBS-A3 Leakage Budget Rule: externally visible responses are minimized until observation intent is classified.

Three Hereditary-Security Rules

- SEC-A1 Split Blueprint Rule: executable lineage information is distributed across incompatible guilds.
- SEC-A2 Living-Context Authorization Rule: edits require spatial, metabolic, and continuity context, not sequence alone.
- SEC-A3 Detached-State Inerting Rule: extracted material remains viable for analysis but cannot reconstruct a body.

Three Rewritten Biological or Engineering Conventions

- Reject permanent differentiation: guild role is reversible within bounded competence.
- Reject one-genome-one-cell completeness: heredity is a threshold archive.
- Reject uniform cellular chemistry: compatibility is achieved at interfaces, not interiors.

Operating Envelopes and Resource Logic

Home range means full function under ordinary resource flow. Refuge range means reduced function, protected cognition, and prioritization of escape or stable dormancy. Death line means continuity can no longer be defended without outside recovery or a separately labeled canon mechanism. Exact temperatures, pressures, doses, power levels, and durations are intentionally left as model parameters until body mass, surface area, mission profile, materials, heat sinks, and energy stores are specified.

Ten-Domain Resilience Profile

Atmospheric Independence and Compound-Event Result

In an inert-gas displacement scenario, the design first stops high-demand activity, isolates gas-exchange assumptions, shifts to finite internal or non-oxygen pathways where its architecture permits, and treats heat rejection as independent from oxygen availability. The design does not

claim indefinite operation. A combined inert-gas, energy-denial, and blocked-heat-rejection case ends when reserves or thermal capacity saturate; this is explicitly a refuge-duration problem rather than “immunity.”

Captivity and Identity Analysis

The design assumes a captor can study materials, observe behavior, collect samples, deny communications, and offer deceptive rescue. The protected objective is not secrecy of composition but prevention of unauthorized developmental authority, identity instantiation, and control. Extracted material is either incomplete, role-limited, revocable, or non-executable. Remote peers cannot command the body. A reconstructed body is not automatically declared the same identity.

Simulation Hook

The roadmap simulation should represent this family at four layers: local unit state; neighborhood or guild state; organ/function state; and continuity/governance state. Escalation to higher fidelity occurs when gradients, trust disagreement, damage concentration, or phase transitions exceed error bounds. Required state variables include energy reserve, heat capacity, repair feedstock, trust status, lineage state, role allocation, signal confidence, and continuity quorum.

Falsification and Principal Failure Modes

- Coordination latency during simultaneous multi-region injury
- Guild-interface corruption that passes local checks
- Loss of enough rare guilds to prevent blueprint reconstruction

Central falsification criterion: reject the family if its defining wager fails to improve survival or continuity relative to a simpler homogeneous baseline under comparable energy, mass, and control assumptions.

Gen-2 Path

Gen-2 work should target the first failed resource, add independent verification around the highest-risk trust boundary, and test whether subsystem benefits survive scale-up. No Gen-2 claim should be made until analytical budgets and digital-twin sensitivity studies identify dominant parameters.

Architecture B: Ceramic Lattice

Core wager: Mechanical continuity and radiation shielding outweigh humanlike softness. Principal sacrifice: reduced agility, slower repair, and high mineral feedstock demand.

Architecture and Omni-cell Model

Substrate: mineral-organic microcells embedded in a crack-arresting ceramic composite. Matrix: graded lattice with sacrificial outer courses and protected internal computation vaults. The design treats omniscience at the cell level as a liability. Universal competence is latent and constrained by local role, environmental state, authorization, and resource availability.

Hereditary, Energy, Thermal, Repair, Identity, and Link Systems

Three Observation Rules

- OBS-B1 Strain-Is-Signal Rule: structural deformation is sensed as a whole-body field.
- OBS-B2 Calibration Witness Rule: every critical sensor is paired with an independent physical standard.

- OBS-B3 Quiet Surface Rule: external scans receive only passive material response unless consent is authenticated.

Three Hereditary-Security Rules

- SEC-B1 Write-Once Ancestral Core: baseline heredity cannot be remotely rewritten.
- SEC-B2 Layered Edit Gate: mutable layers require independent mechanical and cryptographic proof.
- SEC-B3 Sample Dead-End Rule: removed chips lack developmental gradients and cannot self-assemble.

Three Rewritten Biological or Engineering Conventions

- Reject membrane-only cell boundaries: functional units may share solid interfaces.
- Reject soft extracellular matrix as primary structure: the lattice carries load.
- Reject continuous growth morphology: expansion occurs by audited module insertion.

Operating Envelopes and Resource Logic

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Falsification and Principal Failure Modes

- Brittle cascade if microcrack sensing is blinded
- Thermal fin immobilization in vacuum
- Mineral depletion after repeated large-volume repair

Central falsification criterion: reject the family if its defining wager fails to improve survival or continuity relative to a simpler homogeneous baseline under comparable energy, mass, and control assumptions.

Gen-2 Path

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Architecture C: Mycelial Relay

Core wager: Survival depends on preserving connected capability across separable nodes. Principal sacrifice: ambiguous bodily boundaries and difficult ownership/fragmentation governance.

Architecture and Omni-cell Model

Substrate: fibrous biohybrid nodes linked by contractile, conductive, fluid-carrying strands. Matrix: redundant mesh with temporary organs assembled near demand. The design treats omniscience at the cell level as a liability. Universal competence is latent and constrained by local role, environmental state, authorization, and resource availability.

Hereditary, Energy, Thermal, Repair, Identity, and Link Systems

Three Observation Rules

- OBS-C1 Parallax Rule: reality claims require geographically separated nodes where available.
- OBS-C2 Severance Honesty Rule: isolated nodes mark their models as local and possibly stale.
- OBS-C3 Observer Gradient Rule: the body detects and records how external probing changes local behavior.

Three Hereditary-Security Rules

- SEC-C1 Role-Limited Genome Rule: each node knows only its developmental contract.
- SEC-C2 Continuity-Ticket Rule: regeneration of cognition requires a live continuity ticket, not tissue alone.
- SEC-C3 Quarantine Bud Rule: suspicious fragments form sterile diagnostic buds instead of rejoining.

Three Rewritten Biological or Engineering Conventions

- Reject fixed organs: organs are temporary network condensations.
- Reject singular body boundary: embodiment is a governed connected graph.
- Reject inheritance by complete cellular copy: lineage is role-partitioned.

Operating Envelopes and Resource Logic

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Ten-Domain Resilience Profile

Atmospheric Independence and Compound-Event Result

In an inert-gas displacement scenario, the design first stops high-demand activity, isolates gas-exchange assumptions, shifts to finite internal or non-oxygen pathways where its architecture permits, and treats heat rejection as independent from oxygen availability. The design does not claim indefinite operation. A combined inert-gas, energy-denial, and blocked-heat-rejection case ends when reserves or thermal capacity saturate; this is explicitly a refuge-duration problem rather than “immunity.”

Captivity and Identity Analysis

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Simulation Hook

The roadmap simulation should represent this family at four layers: local unit state; neighborhood or guild state; organ/function state; and continuity/governance state. Escalation to higher fidelity occurs when gradients, trust disagreement, damage concentration, or phase transitions exceed error bounds. Required state variables include energy reserve, heat capacity, repair feedstock, trust status, lineage state, role allocation, signal confidence, and continuity quorum.

Falsification and Principal Failure Modes

- Consensus split after long partition
- Adversary-controlled bridge node
- High water and nutrient demand during rapid regrowth

Central falsification criterion: reject the family if its defining wager fails to improve survival or continuity relative to a simpler homogeneous baseline under comparable energy, mass, and control assumptions.

Gen-2 Path

Gen-2 work should target the first failed resource, add independent verification around the highest-risk trust boundary, and test whether subsystem benefits survive scale-up. No Gen-2 claim should be made until analytical budgets and digital-twin sensitivity studies identify dominant parameters.

Architecture D: Phase-Core Mantle

Core wager: The first failure in extreme environments is often heat, solvent, or phase instability. Principal sacrifice: mass, acceleration, and high-output endurance.

Architecture and Omni-cell Model

Substrate: hydrogel microcells surrounding phase-change capsules and catalytic microreactors. Matrix: concentric mantle zones around a low-power continuity core. The design treats omniscience at the cell level as a liability. Universal competence is latent and constrained by local role, environmental state, authorization, and resource availability.

Hereditary, Energy, Thermal, Repair, Identity, and Link Systems

Three Observation Rules

- OBS-D1 Thermal Truth Rule: sensor confidence is discounted during rapid phase transitions.
- OBS-D2 Minimum-Cognition Rule: refuge mode preserves only observations needed for continuity and escape.
- OBS-D3 Signature Conservation Rule: emitted heat and chemistry are treated as information leakage.

Three Hereditary-Security Rules

- SEC-D1 Multi-Phase Key Rule: edit authority exists only when multiple physical phases are present.
- SEC-D2 Dormant Archive Rule: cold archives remain non-executable until continuity proof is restored.
- SEC-D3 Sample Dissociation Rule: removed gel loses the gradients required for organized development.

Three Rewritten Biological or Engineering Conventions

- Reject constant metabolic activity: prolonged hypometabolism is normal operation.
- Reject uniform body temperature: controlled regional temperature differences are functional.
- Reject fixed cellular water content: cells shift among hydrated, glassy, and gel states.

Operating Envelopes and Resource Logic

Home range means full function under ordinary resource flow. Refuge range means reduced function, protected cognition, and prioritization of escape or stable dormancy. Death line means continuity can no longer be defended without outside recovery or a separately labeled canon mechanism. Exact temperatures, pressures, doses, power levels, and durations are intentionally left as model parameters until body mass, surface area, mission profile, materials, heat sinks, and energy stores are specified.

Ten-Domain Resilience Profile

Atmospheric Independence and Compound-Event Result

In an inert-gas displacement scenario, the design first stops high-demand activity, isolates gas-exchange assumptions, shifts to finite internal or non-oxygen pathways where its architecture permits, and treats heat rejection as independent from oxygen availability. The design does not claim indefinite operation. A combined inert-gas, energy-denial, and blocked-heat-rejection case ends when reserves or thermal capacity saturate; this is explicitly a refuge-duration problem rather than “immunity.”

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Simulation Hook

The roadmap simulation should represent this family at four layers: local unit state; neighborhood or guild state; organ/function state; and continuity/governance state. Escalation to higher fidelity occurs when gradients, trust disagreement, damage concentration, or phase transitions exceed

error bounds. Required state variables include energy reserve, heat capacity, repair feedstock, trust status, lineage state, role allocation, signal confidence, and continuity quorum.

Falsification and Principal Failure Modes

- Finite phase-change capacity in blocked heat rejection
- Rewarming injury after deep refuge
- Low performance during abrupt high-power demand

Central falsification criterion: reject the family if its defining wager fails to improve survival or continuity relative to a simpler homogeneous baseline under comparable energy, mass, and control assumptions.

Gen-2 Path

Gen-2 work should target the first failed resource, add independent verification around the highest-risk trust boundary, and test whether subsystem benefits survive scale-up. No Gen-2 claim should be made until analytical budgets and digital-twin sensitivity studies identify dominant parameters.

Architecture E: Archive Swarm

Core wager: Capture and destruction are harder when embodiment is physically distributed. Principal sacrifice: weaker local strength and difficult continuity governance.

Architecture and Omni-cell Model

Substrate: millimeter-to-centimeter modular microbodies with restricted specialties. Matrix: cooperating swarm that can form temporary tissues and structures. The design treats omniscience at the cell level as a liability. Universal competence is latent and constrained by local role, environmental state, authorization, and resource availability.

Hereditary, Energy, Thermal, Repair, Identity, and Link Systems

Three Observation Rules

- OBS-E1 Population Evidence Rule: no single module can define external reality.
- OBS-E2 Byzantine Sensing Rule: models tolerate malicious or defective observers.
- OBS-E3 Sampling Ambiguity Rule: modules reveal only role-specific behavior under examination.

Three Hereditary-Security Rules

- SEC-E1 Erasure-Coded Lineage Rule: reconstruction requires a threshold of diverse modules.
- SEC-E2 Foundry Consent Rule: new identity-bearing modules require live quorum authorization.
- SEC-E3 Captured-Module Revocation Rule: compromised modules lose trust without being destroyed.

Three Rewritten Biological or Engineering Conventions

- Reject one body-one organism mapping: embodiment is a mobile collective.
- Reject centralized development: modules are fabricated through distributed foundries.
- Reject persistent organ topology: organ function is a temporary swarm formation.

Operating Envelopes and Resource Logic

Home range means full function under ordinary resource flow. Refuge range means reduced function, protected cognition, and prioritization of escape or stable dormancy. Death line means

continuity can no longer be defended without outside recovery or a separately labeled canon mechanism. Exact temperatures, pressures, doses, power levels, and durations are intentionally left as model parameters until body mass, surface area, mission profile, materials, heat sinks, and energy stores are specified.

Ten-Domain Resilience Profile

Atmospheric Independence and Compound-Event Result

In an inert-gas displacement scenario, the design first stops high-demand activity, isolates gas-exchange assumptions, shifts to finite internal or non-oxygen pathways where its architecture permits, and treats heat rejection as independent from oxygen availability. The design does not claim indefinite operation. A combined inert-gas, energy-denial, and blocked-heat-rejection case ends when reserves or thermal capacity saturate; this is explicitly a refuge-duration problem rather than “immunity.”

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Falsification and Principal Failure Modes

- Quorum capture through gradual infiltration
- Identity ambiguity when two large partitions evolve independently
- Loss of coordination in high-jamming environments

Central falsification criterion: reject the family if its defining wager fails to improve survival or continuity relative to a simpler homogeneous baseline under comparable energy, mass, and control assumptions.

Gen-2 Path

Gen-2 work should target the first failed resource, add independent verification around the highest-risk trust boundary, and test whether subsystem benefits survive scale-up. No Gen-2 claim should be made until analytical budgets and digital-twin sensitivity studies identify dominant parameters.

Architecture F: Electrogenic Reef

Core wager: Long service life requires harvesting many weak energy sources continuously. Principal sacrifice: visibility, environmental coupling, and dependence on conductive media or installed surfaces.

Architecture and Omni-cell Model

Substrate: electrogenic biohybrid cells coupled to conductive polymer vasculature. Matrix: branched reef-like energy and computation surfaces. The design treats omniscience at the cell level as a liability. Universal competence is latent and constrained by local role, environmental state, authorization, and resource availability.

Hereditary, Energy, Thermal, Repair, Identity, and Link Systems

Three Observation Rules

- OBS-F1 Energy-Correlated Sensing Rule: sensor reports include the energy source and interference state.
- OBS-F2 Environmental Reciprocity Rule: observations estimate how the body is altering the environment.
- OBS-F3 Emission Ledger Rule: power harvesting and communication signatures are continuously audited.

Three Hereditary-Security Rules

- SEC-F1 Dual-Substrate Authority Rule: DNA cannot authorize control-lattice edits by itself.
- SEC-F2 Energy-Provenance Rule: suspicious energy interfaces have no update authority.
- SEC-F3 Cultured-Tissue Limit Rule: maintenance cells cannot instantiate cognition or lineage control.

Three Rewritten Biological or Engineering Conventions

- Reject food-only metabolism: ambient gradients are legitimate primary inputs.
- Reject closed circulation: some exchange surfaces are permanently environmental.
- Reject genetic monopoly on control: critical governance resides in a non-genetic lattice.

Operating Envelopes and Resource Logic

Home range means full function under ordinary resource flow. Refuge range means reduced function, protected cognition, and prioritization of escape or stable dormancy. Death line means continuity can no longer be defended without outside recovery or a separately labeled canon mechanism. Exact temperatures, pressures, doses, power levels, and durations are intentionally left as model parameters until body mass, surface area, mission profile, materials, heat sinks, and energy stores are specified.

Ten-Domain Resilience Profile

Atmospheric Independence and Compound-Event Result

In an inert-gas displacement scenario, the design first stops high-demand activity, isolates gas-exchange assumptions, shifts to finite internal or non-oxygen pathways where its architecture permits, and treats heat rejection as independent from oxygen availability. The design does not claim indefinite operation. A combined inert-gas, energy-denial, and blocked-heat-rejection case ends when reserves or thermal capacity saturate; this is explicitly a refuge-duration problem rather than “immunity.”

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Falsification and Principal Failure Modes

- Energy-starved sterile environment
- Conductive network corrosion or poisoning
- Detection through persistent harvesting signatures

Central falsification criterion: reject the family if its defining wager fails to improve survival or continuity relative to a simpler homogeneous baseline under comparable energy, mass, and control assumptions.

Gen-2 Path

Gen-2 work should target the first failed resource, add independent verification around the highest-risk trust boundary, and test whether subsystem benefits survive scale-up. No Gen-2 claim should be made until analytical budgets and digital-twin sensitivity studies identify dominant parameters.

Architecture G: Chrysalis Ledger

Core wager: Survival is sometimes the ability to stop safely before resources become fatal. Principal sacrifice: reduced availability and delayed response during refuge.

Architecture and Omni-cell Model

Substrate: conventional biohybrid cells optimized for reversible vitrified refuge states. Matrix: human-scale adaptive body with distributed refuge capsules. The design treats omniscience at the cell level as a liability. Universal competence is latent and constrained by local role, environmental state, authorization, and resource availability.

Hereditary, Energy, Thermal, Repair, Identity, and Link Systems

Three Observation Rules

- OBS-G1 Wake Skepticism Rule: post-dormancy observations begin untrusted.
- OBS-G2 Continuity Trace Rule: internal state transitions are logged redundantly before shutdown.
- OBS-G3 Minimal Exposure Rule: refuge mode emits no active diagnostic response absent authorization.

Three Hereditary-Security Rules

- SEC-G1 Wake-Key Diversity Rule: awakening requires independent biological, temporal, and continuity proofs.
- SEC-G2 Reference Genome Separation Rule: dormant reference copies cannot initiate development.
- SEC-G3 Recovery Sandbox Rule: repaired tissue operates in restricted mode before rejoining.

Three Rewritten Biological or Engineering Conventions

- Reject continuous consciousness as a survival requirement: governed dormancy is permitted.

- Reject continuous circulation: refuge state uses immobilized protected compartments.
- Reject immediate regeneration: repair waits for authenticated recovery conditions.

Operating Envelopes and Resource Logic

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Falsification and Principal Failure Modes

- Wake failure after long-term chemical drift
- Adversarial false wake environment
- Vitrification or rewarming damage at large scale

Central falsification criterion: reject the family if its defining wager fails to improve survival or continuity relative to a simpler homogeneous baseline under comparable energy, mass, and control assumptions.

Gen-2 Path

Gen-2 work should target the first failed resource, add independent verification around the highest-risk trust boundary, and test whether subsystem benefits survive scale-up. No Gen-2 claim should be made until analytical budgets and digital-twin sensitivity studies identify dominant parameters.

Architecture H: Axiom Shell

Core wager: Some project requirements exceed known physics and should be isolated behind declared axioms. Principal sacrifice: physical-track performance is modest; canon capabilities depend on Architect-approved laws.

Architecture and Omni-cell Model

Substrate: physical biohybrid shell plus a separately defined canon continuity anchor. Matrix: conventional distributed body with interfaces reserved for canon-layer functions. The design treats omniscience at the cell level as a liability. Universal competence is latent and constrained by local role, environmental state, authorization, and resource availability.

Hereditary, Energy, Thermal, Repair, Identity, and Link Systems

Three Observation Rules

- OBS-H1 Track Separation Rule: an observation must state whether it is physical, simulated, or canon-derived.
- OBS-H2 Causality Rule: physical conclusions cannot use information received outside causal limits.
- OBS-H3 Axiom Challenge Rule: canon observations must identify the axiom permitting them and its failure condition.

Three Hereditary-Security Rules

- SEC-H1 Cross-Track Non-Escalation Rule: canon credentials do not automatically modify biology.
- SEC-H2 Physical Root Rule: bodily edits require physically present authenticated authority.
- SEC-H3 Canon Quarantine Rule: ontological claims are isolated until independently reconciled.

Three Rewritten Biological or Engineering Conventions

- Reject a single evidence ontology: physical and canon truth claims remain separate.
- Reject instantaneous communication in the physical track: latency is unavoidable.
- Reject restoration-equals-survival: continuity status remains explicitly unresolved.

Operating Envelopes and Resource Logic

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Ten-Domain Resilience Profile

Atmospheric Independence and Compound-Event Result

In an inert-gas displacement scenario, the design first stops high-demand activity, isolates gas-exchange assumptions, shifts to finite internal or non-oxygen pathways where its architecture permits, and treats heat rejection as independent from oxygen availability. The design does not claim indefinite operation. A combined inert-gas, energy-denial, and blocked-heat-rejection case ends when reserves or thermal capacity saturate; this is explicitly a refuge-duration problem rather than “immunity.”

Captivity and Identity Analysis

The design assumes a captor can study materials, observe behavior, collect samples, deny communications, and offer deceptive rescue. The protected objective is not secrecy of composition but prevention of unauthorized developmental authority, identity instantiation, and control. Extracted material is either incomplete, role-limited, revocable, or non-executable. Remote peers cannot command the body. A reconstructed body is not automatically declared the same identity.

Simulation Hook

The roadmap simulation should represent this family at four layers: local unit state; neighborhood or guild state; organ/function state; and continuity/governance state. Escalation to higher fidelity occurs when gradients, trust disagreement, damage concentration, or phase transitions exceed error bounds. Required state variables include energy reserve, heat capacity, repair feedstock, trust status, lineage state, role allocation, signal confidence, and continuity quorum.

Falsification and Principal Failure Modes

- Canon interface becomes a narrative loophole
- Physical shell remains vulnerable without canon support
- Continuity dispute after reconstruction

Central falsification criterion: reject the family if its defining wager fails to improve survival or continuity relative to a simpler homogeneous baseline under comparable energy, mass, and control assumptions.

Gen-2 Path

Gen-2 work should target the first failed resource, add independent verification around the highest-risk trust boundary, and test whether subsystem benefits survive scale-up. No Gen-2 claim should be made until analytical budgets and digital-twin sensitivity studies identify dominant parameters.

1AI / -1AI Constructive-Adversarial Cycle

The constructive pass maximized each family around its core wager. The adversarial pass assumed blueprint knowledge, long preparation time, institutional resources, and preference for intact capture or identity compromise. The following attacks succeeded conceptually and remain visible rather than being patched away by assertion.

Cross-Family Common-Mode Failures

- All physical families ultimately require a heat sink or finite thermal storage.
- All repair systems require matter, energy, time, and trustworthy state information.
- All identity systems depend on a continuity rule that remains partly philosophical and governance-based.
- All long-distance links can be delayed, jammed, observed, corrupted, or partitioned.
- All authorization systems can fail through compromised roots, coerced consent, or stale state.
- All universal-competence systems risk common-mode exploitation unless heterogeneity is preserved.
- All environmental survivability claims collapse if simultaneous stressors saturate the same reserve.

Pareto Analysis and Portfolio Selection

The evaluation does not average all criteria into one “best” body. Different architectures dominate under different mission assumptions. Mosaic Covenant offers the strongest balanced physical-track starting point because heterogeneity addresses monoculture and sample-theft risk without requiring total physical distribution. Chrysalis Ledger is the strongest reserve because governed dormancy can protect continuity when active performance is impossible. Archive Swarm is the wildcard because it changes the body boundary and may resist single-site capture, but it creates the hardest identity and quorum problems. Phase-Core Mantle should donate thermal and refuge subsystems to other families. Axiom Shell should govern canon integration rather than compete as the primary physical platform.

Next-Step Recommendations

4. Lock the eight family definitions before reviewing other AIs’ prototype sets.
5. Build low-order analytical models for energy, heat, repair feedstock, and trust-state dynamics before assigning exact performance numbers.
6. Select three mission profiles, terrestrial hostile confinement, deep-space transit, and long-duration isolated field operation, then test the same families under each.
7. Develop an explicit Continuity Standard deciding when repair, dormancy, fragmentation, restoration, and duplication count as survival.
8. Establish separate SIM, MODEL, COMPONENT, INTEGRATED, INCIDENT, HOSTILE, and CANON evidence ledgers.
9. Prototype the interoperability layer as a capability-limited, revocable, delay-tolerant protocol independent of any one family.
10. Use Mosaic Covenant, Chrysalis Ledger, Archive Swarm, and Phase-Core Mantle as the first digital portfolio.
11. Keep biological work non-operational until independent ethics, ecological containment, and biosecurity review are complete.

Questions Requiring Architect Decision

- Is the 4,712 figure a locked narrative fact while the taxonomy remains draft, or is the count itself still under review?
- Does continuity require uninterrupted cognition, or may governed dormancy preserve the same entity?
- Can two separated fragments both retain legitimate continuity, or must one become a successor?
- Are destructive responses affecting detached samples categorically prohibited, or only excluded from the physical-track baseline?
- How much human compatibility, appearance, and social interoperability is required?
- May the body deliberately distribute itself across multiple physical units?
- Which canon axioms, if any, may govern ONT threats and cross-dimensional linkage?
- What minimum standard converts a simulated survival result into a validated project claim?

Comments and Concerns

The largest technical risk is not one dramatic extinction event. It is compound resource exhaustion: damage consumes repair feedstock; repair raises energy demand; energy becomes heat; heat rejection is blocked; cognition loses sensor integrity; security decisions become unreliable. The simulation program should prioritize these cascades over isolated headline scenarios.

The largest governance risk is identity ambiguity. A physiology can survive biologically while losing autonomy, continuity, or ownership of its own developmental process. The project should not defer continuity rules until after a restoration or fragmentation mechanism has already been canonized.

The largest research integrity risk is allowing canon mechanisms to inherit credibility from real biological motifs. Tardigrade proteins, hypometabolic states, synthetic cells, and distributed biological computation are useful inspirations, but none demonstrates the proposed whole-body systems.

Visual Appendix: Charts, Graphs, Tables, Rows, Images, Matrices, and Diagrams

All values in conceptual score charts are design-assessment targets generated for comparison. They are not experimental measurements.

Exhibit V1. Eight-Family Conceptual Comparison

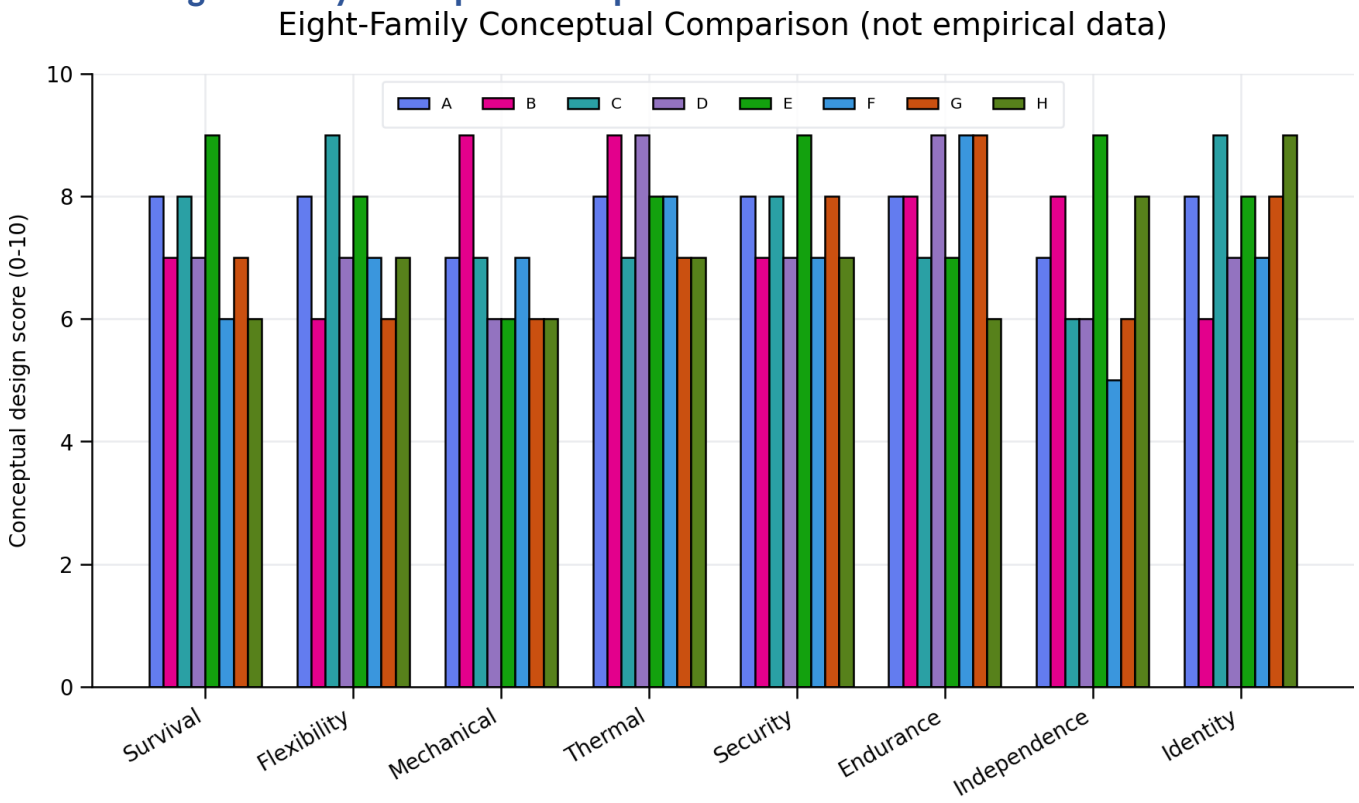


Exhibit V2. Endurance-Security Pareto View

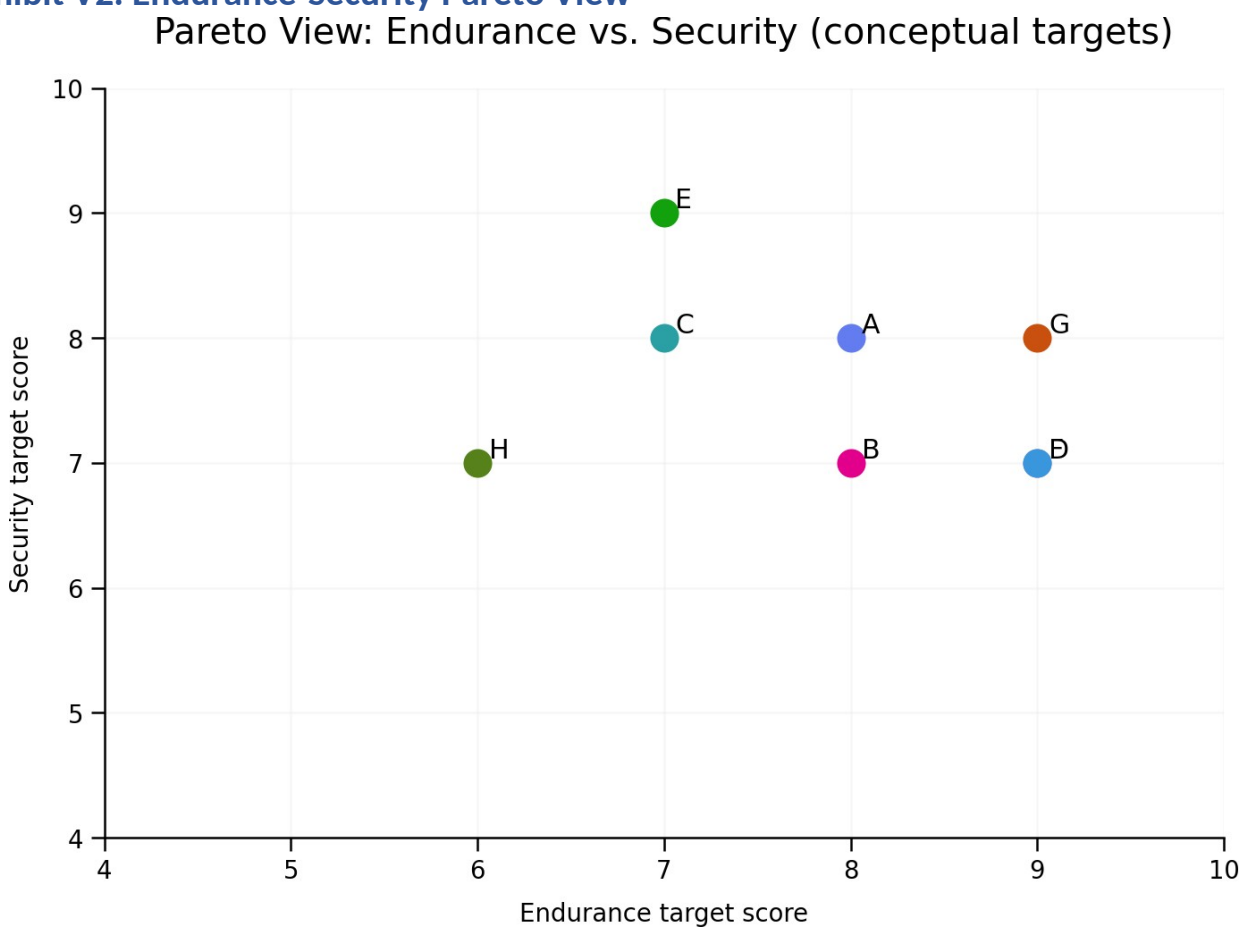


Exhibit V3. Orthogonality Matrix

Exhibit V4. Evidence Classes

Exhibit V5. Continuity State Model

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Exhibit V6. Full Self-Attack Ledger

Exhibit V7. Ten-Domain Evaluation Frame

References

Version and Independence Lock

Microsoft 365 Copilot Independent Architecture Set v1.0 is version-locked as of October 28, 2026. The eight families above were generated as one internal portfolio before reviewing any completed prototype sets from other collaborating AIs. Later comparison may identify convergence, but this document does not claim uniqueness relative to unseen work.

Supplemental Visual Appendix: Relocated Main-Body Tables

Revision v1.1 relocates every substantive table, row/column matrix, chart, and visual representation out of the narrative body. The only table retained before the appendix is the cover-page document-control block. The following exhibits preserve the information from the removed tables for internal research use.

Exhibit R1. Relocated Table

Architectural thesis
My Omnicellular Physiology is fundamentally built around governed heterogeneity and recoverable continuity rather than one universal cell type, one central organ topology, or an assumption of invulnerability.

Exhibit R2. Relocated Table

Mission
Design and honestly assess eight materially different Gen-1 embodiment families that preserve autonomy, continuity, repairability, and environmental survival without hiding costs, uncertainty, or dependence behind science-fiction vocabulary.

Exhibit R3. Relocated Table

Issue	Finding	Executed amendment
Ambiguity	“Withstand” could mean function, survival, dormancy, escape, or restoration.	Each threat report names the exact outcome.
Contradiction	The 4,712 framework is marked draft while calling the number locked; it also conflicts on simulation counting.	Treat taxonomy as project reference; keep evidence ledgers separate.
Unsupported absolute	Cross-dimensional, zero-latency, entropy-exempt, or perfect recycling claims exceed established physics.	Confine to canon track with replacement axioms.
Quantification gap	Body mass, mission time, power, feedstocks, and heat sinks are unspecified.	Use ranges or qualitative budgets; label exact numbers unknown.
Identity gap	Repair, duplicate, fragment, and restoration are not equivalent.	Use explicit continuity states and Architect decisions.
Security risk	Blueprint secrecy cannot be assumed.	Use living context, threshold authorization, compartmentalization, and revocation.
Safety concern	Hostile scenarios could become harmful protocols.	Use conceptual stressors only; no operational exposure or bioengineering directions.
Execution volume	Eight families x all tests is large.	Narrative dossiers plus visual appendices and standardized ledgers.

Exhibit R4. Relocated Table

Source	Class	Used for	Control
OMNICELLULAR_Physiology_Independent_Research_Directive_v2.0.docx	Controlling directive	Requirements and locked decisions	Not scientific evidence
The_Omnicellular_Vision_Wattpad.md	Narrative/vision	1AI builder and -1AI challenger; scale ambition	Canon-style claims need validation
Omnicellular_Architecture_15_Year_Roadmap.md	Strategic roadmap	Multi-scale abstraction; staged 2026-2041 program	Budgets, accuracy, and timelines are targets
Omnicellular_Architecture_Notes.md	Concept notes/mixed research	Omni-cell ambition and population scale	Embedded citations and estimates require source review

4,712 Extinction Attempt Classification Framework.docx	Draft taxonomy/canon reference	Ten domains and four severity tiers	Draft/locked tension; counting contradiction
Public literature	Scientific context	Telomeres, hypoxia, synthetic cells, radiation, heat shock, distributed systems	Does not validate proposed whole-body architectures

Exhibit R5. Relocated Table

Working definition
An Omni-cell is a bounded biological, artificial, or biohybrid unit with broad latent competence, a shared interoperability interface, and governed ability to assume multiple roles. No requirement is imposed that every unit perform every function simultaneously or hold a complete executable blueprint.

Exhibit R6. Relocated Table

Architectural thesis
My Omnicellular Physiology is fundamentally built around distributed heterogeneity rather than universal cellular sameness.

Exhibit R7. Relocated Table

Domain	Design
Hereditary substrate	permissioned DNA/XNA split archive; no cell holds a complete executable blueprint
Energy/respiration	aerobic metabolism in normal operation, fermentation and redox-reserve chemistry in refuge operation
Thermal regulation	vascular microloops, phase-change vesicles, surface radiators, and enforced metabolic throttling
Repair	local guild replacement followed by cross-guild verification
Identity continuity	continuity quorum distributed across cognition-bearing guilds; no single cell is the identity root
Long-distance link	delay-tolerant authenticated store-and-forward; no remote command authority

Exhibit R8. Relocated Table

Domain	Gen-1 response
ENV-T	Active regulation in Home Range; throttling, buffering, or dispersal in Refuge; finite heat-sink saturation remains the death line.
ENV-A	Normal respiration changes to reserve or suppressed metabolism; vacuum requires structural and thermal protection, not merely oxygen independence.
ENV-R	Shielding, redundancy, damage sensing, and repair reduce risk; no claim of absolute immunity.
BIO	Interface filtering, heterogeneity, quarantine, and lineage checks reduce common-mode infection; unknown chemistry remains a risk.
MECH	Distributed structure and repair limit local damage; sufficiently rapid total fragmentation can exceed continuity mechanisms.
DEP	Energy and matter reserves support refuge; waste heat and missing repair feedstock remain limiting.
ENE	Surge isolation and segmented conductive paths reduce electrical or plasma damage; high energy can still overwhelm local materials.
INFO	Diverse sensing, bounded updates, and continuity checks

	resist spoofing; coherent corruption across all trust roots is critical.
ONT	No physical-track defense. Canon track requires an explicit replacement axiom.
CIV	Offline autonomy and distributed knowledge improve survival; planetary resource loss may force dormancy, migration, or failure.

Exhibit R9. Relocated Table

Architectural thesis **My Omnicellular Physiology is fundamentally built around load-bearing computational material rather than soft-tissue anatomy.**

Exhibit R10. Relocated Table

Domain	Design
Hereditary substrate	write-once reference genome plus mutable operational epigenome
Energy/respiration	electrochemical cells, environmental thermoelectric capture, and replaceable solid-state reserve blocks
Thermal regulation	high-conductivity heat trunks, reversible heat-storage salts, deployable radiative fins
Repair	crack mapping, local dissolution, additive mineral deposition, and integrity proof
Identity continuity	protected distributed state vaults with continuous cross-checking
Long-distance link	optical/radio relay with physically separated trust modules

Exhibit R11. Relocated Table

Domain	Gen-1 response
ENV-T	Active regulation in Home Range; throttling, buffering, or dispersal in Refuge; finite heat-sink saturation remains the death line.
ENV-A	Normal respiration changes to reserve or suppressed metabolism; vacuum requires structural and thermal protection, not merely oxygen independence.
ENV-R	Shielding, redundancy, damage sensing, and repair reduce risk; no claim of absolute immunity.
BIO	Interface filtering, heterogeneity, quarantine, and lineage checks reduce common-mode infection; unknown chemistry remains a risk.
MECH	Distributed structure and repair limit local damage; sufficiently rapid total fragmentation can exceed continuity mechanisms.
DEP	Energy and matter reserves support refuge; waste heat and missing repair feedstock remain limiting.
ENE	Surge isolation and segmented conductive paths reduce electrical or plasma damage; high energy can still overwhelm local materials.
INFO	Diverse sensing, bounded updates, and continuity checks resist spoofing; coherent corruption across all trust roots is critical.
ONT	No physical-track defense. Canon track requires an explicit replacement axiom.
CIV	Offline autonomy and distributed knowledge improve survival; planetary resource loss may force dormancy, migration, or

	failure.

Exhibit R12. Relocated Table

Architectural thesis
My Omnicellular Physiology is fundamentally built around regenerative network topology rather than a single bounded body.

Exhibit R13. Relocated Table

Domain	Design
Hereditary substrate	node-specific partial genomes with signed developmental roles
Energy/respiration	distributed nutrient oxidation, microbial electrosynthesis analogues, and environmental scavenging
Thermal regulation	wide-area heat spreading and selective node shedding
Repair	rapid rerouting, strand regrowth, and temporary organ budding
Identity continuity	single continuity process maintained by synchronized but non-sovereign cognition nodes
Long-distance link	short-range physical mesh plus long-range asynchronous consensus messages

Exhibit R14. Relocated Table

Domain	Gen-1 response
ENV-T	Active regulation in Home Range; throttling, buffering, or dispersal in Refuge; finite heat-sink saturation remains the death line.
ENV-A	Normal respiration changes to reserve or suppressed metabolism; vacuum requires structural and thermal protection, not merely oxygen independence.
ENV-R	Shielding, redundancy, damage sensing, and repair reduce risk; no claim of absolute immunity.
BIO	Interface filtering, heterogeneity, quarantine, and lineage checks reduce common-mode infection; unknown chemistry remains a risk.
MECH	Distributed structure and repair limit local damage; sufficiently rapid total fragmentation can exceed continuity mechanisms.
DEP	Energy and matter reserves support refuge; waste heat and missing repair feedstock remain limiting.
ENE	Surge isolation and segmented conductive paths reduce electrical or plasma damage; high energy can still overwhelm local materials.
INFO	Diverse sensing, bounded updates, and continuity checks resist spoofing; coherent corruption across all trust roots is critical.
ONT	No physical-track defense. Canon track requires an explicit replacement axiom.
CIV	Offline autonomy and distributed knowledge improve survival; planetary resource loss may force dormancy, migration, or failure.

Exhibit R15. Relocated Table

Architectural thesis

My Omnicellular Physiology is fundamentally built around thermal and chemical buffering rather than peak metabolic performance.

Exhibit R16. Relocated Table

Domain	Design
Hereditary substrate	RNA/DNA hybrid archive with cold, heat, and radiation copies in separate zones
Energy/respiration	slow oxidative metabolism, reversible chemical storage, and deep metabolic suppression
Thermal regulation	latent-heat buffers, evaporative exchange when possible, radiation surfaces, and refuge shutdown
Repair	gel reflow, capsule exchange, and low-temperature reconstruction
Identity continuity	continuity core remains active at minimal rate while peripheral cognition degrades gracefully
Long-distance link	bursty communication scheduled around thermal budget

Exhibit R17. Relocated Table

Domain	Gen-1 response
ENV-T	Active regulation in Home Range; throttling, buffering, or dispersal in Refuge; finite heat-sink saturation remains the death line.
ENV-A	Normal respiration changes to reserve or suppressed metabolism; vacuum requires structural and thermal protection, not merely oxygen independence.
ENV-R	Shielding, redundancy, damage sensing, and repair reduce risk; no claim of absolute immunity.
BIO	Interface filtering, heterogeneity, quarantine, and lineage checks reduce common-mode infection; unknown chemistry remains a risk.
MECH	Distributed structure and repair limit local damage; sufficiently rapid total fragmentation can exceed continuity mechanisms.
DEP	Energy and matter reserves support refuge; waste heat and missing repair feedstock remain limiting.
ENE	Surge isolation and segmented conductive paths reduce electrical or plasma damage; high energy can still overwhelm local materials.
INFO	Diverse sensing, bounded updates, and continuity checks resist spoofing; coherent corruption across all trust roots is critical.
ONT	No physical-track defense. Canon track requires an explicit replacement axiom.
CIV	Offline autonomy and distributed knowledge improve survival; planetary resource loss may force dormancy, migration, or failure.

Exhibit R18. Relocated Table

Architectural thesis
My Omnicellular Physiology is fundamentally built around many bounded microbodies rather than one continuous organism.

Exhibit R19. Relocated Table

Domain	Design
Hereditary substrate	erasure-coded lineage shards distributed across many modules
Energy/respiration	mixed solar, chemical, thermal-gradient, and stored electrochemical capture
Thermal regulation	small-module heat rejection, dispersal, shade seeking, and sacrificial thermal zoning
Repair	module replication from authorized foundries and swarm reconstitution
Identity continuity	continuity is the causal process across a moving authenticated quorum, not any module
Long-distance link	local mesh with delayed long-haul summaries; partitions remain autonomous

Exhibit R20. Relocated Table

Domain	Gen-1 response
ENV-T	Active regulation in Home Range; throttling, buffering, or dispersal in Refuge; finite heat-sink saturation remains the death line.
ENV-A	Normal respiration changes to reserve or suppressed metabolism; vacuum requires structural and thermal protection, not merely oxygen independence.
ENV-R	Shielding, redundancy, damage sensing, and repair reduce risk; no claim of absolute immunity.
BIO	Interface filtering, heterogeneity, quarantine, and lineage checks reduce common-mode infection; unknown chemistry remains a risk.
MECH	Distributed structure and repair limit local damage; sufficiently rapid total fragmentation can exceed continuity mechanisms.
DEP	Energy and matter reserves support refuge; waste heat and missing repair feedstock remain limiting.
ENE	Surge isolation and segmented conductive paths reduce electrical or plasma damage; high energy can still overwhelm local materials.
INFO	Diverse sensing, bounded updates, and continuity checks resist spoofing; coherent corruption across all trust roots is critical.
ONT	No physical-track defense. Canon track requires an explicit replacement axiom.
CIV	Offline autonomy and distributed knowledge improve survival; planetary resource loss may force dormancy, migration, or failure.

Exhibit R21. Relocated Table

Architectural thesis **My Omnicellular Physiology is fundamentally built around environmental energy integration rather than sealed metabolic independence.**

Exhibit R22. Relocated Table

Domain	Design
Hereditary substrate	conventional DNA for maintenance cells plus protected non-DNA control lattice

Energy/respiration	redox gradients, microbial fuel analogues, light capture, vibration, and thermoelectric scavenging
Thermal regulation	large conductive surface, fluid circulation, and energy-aware workload placement
Repair	slow accretion, plug-in patches, and self-testing vascular replacements
Identity continuity	a low-power stable core with replaceable peripheral energy organs
Long-distance link	surface antennas and conductive infrastructure with offline local operation

Exhibit R23. Relocated Table

Domain	Gen-1 response
ENV-T	Active regulation in Home Range; throttling, buffering, or dispersal in Refuge; finite heat-sink saturation remains the death line.
ENV-A	Normal respiration changes to reserve or suppressed metabolism; vacuum requires structural and thermal protection, not merely oxygen independence.
ENV-R	Shielding, redundancy, damage sensing, and repair reduce risk; no claim of absolute immunity.
BIO	Interface filtering, heterogeneity, quarantine, and lineage checks reduce common-mode infection; unknown chemistry remains a risk.
MECH	Distributed structure and repair limit local damage; sufficiently rapid total fragmentation can exceed continuity mechanisms.
DEP	Energy and matter reserves support refuge; waste heat and missing repair feedstock remain limiting.
ENE	Surge isolation and segmented conductive paths reduce electrical or plasma damage; high energy can still overwhelm local materials.
INFO	Diverse sensing, bounded updates, and continuity checks resist spoofing; coherent corruption across all trust roots is critical.
ONT	No physical-track defense. Canon track requires an explicit replacement axiom.
CIV	Offline autonomy and distributed knowledge improve survival; planetary resource loss may force dormancy, migration, or failure.

Exhibit R24. Relocated Table

Architectural thesis **My Omnicellular Physiology is fundamentally built around provable dormancy and recovery rather than continuous full function.**

Exhibit R25. Relocated Table

Domain	Design
Hereditary substrate	conventional genome with strong error correction and offline reference copies
Energy/respiration	normal oxidative metabolism with emergency chemical stores and near-zero-power dormancy
Thermal regulation	controlled cooling, vitrification analogues, insulation, and slow radiative balance
Repair	awakening-stage audit before growth or functional restoration

Identity continuity	identity ledger records the causal path into and out of dormancy; backups are not automatically the same entity
Long-distance link	no link required during refuge; wake messages are advisory, not authoritative

Exhibit R26. Relocated Table

Domain	Gen-1 response
ENV-T	Active regulation in Home Range; throttling, buffering, or dispersal in Refuge; finite heat-sink saturation remains the death line.
ENV-A	Normal respiration changes to reserve or suppressed metabolism; vacuum requires structural and thermal protection, not merely oxygen independence.
ENV-R	Shielding, redundancy, damage sensing, and repair reduce risk; no claim of absolute immunity.
BIO	Interface filtering, heterogeneity, quarantine, and lineage checks reduce common-mode infection; unknown chemistry remains a risk.
MECH	Distributed structure and repair limit local damage; sufficiently rapid total fragmentation can exceed continuity mechanisms.
DEP	Energy and matter reserves support refuge; waste heat and missing repair feedstock remain limiting.
ENE	Surge isolation and segmented conductive paths reduce electrical or plasma damage; high energy can still overwhelm local materials.
INFO	Diverse sensing, bounded updates, and continuity checks resist spoofing; coherent corruption across all trust roots is critical.
ONT	No physical-track defense. Canon track requires an explicit replacement axiom.
CIV	Offline autonomy and distributed knowledge improve survival; planetary resource loss may force dormancy, migration, or failure.

Exhibit R27. Relocated Table

Architectural thesis **My Omnicellular Physiology is fundamentally built around explicit dual-track embodiment rather than disguising canon mechanisms as engineering.**

Exhibit R28. Relocated Table

Domain	Design
Hereditary substrate	physical threshold genome; canon layer stores no directly executable biological sequence
Energy/respiration	physical metabolism and stored energy; canon energy claims are prohibited unless separately axiomatized
Thermal regulation	ordinary conductive, convective, radiative, and phase-change controls
Repair	physical repair where possible; canon restoration is labeled successor/duplicate until continuity is decided
Identity continuity	physical continuity is primary; any ontological anchor is an unresolved Architect decision
Long-distance link	physical signaling obeys causal delay; canon linkage requires

	a replacement axiom

Exhibit R29. Relocated Table

Domain	Gen-1 response
ENV-T	Active regulation in Home Range; throttling, buffering, or dispersal in Refuge; finite heat-sink saturation remains the death line.
ENV-A	Normal respiration changes to reserve or suppressed metabolism; vacuum requires structural and thermal protection, not merely oxygen independence.
ENV-R	Shielding, redundancy, damage sensing, and repair reduce risk; no claim of absolute immunity.
BIO	Interface filtering, heterogeneity, quarantine, and lineage checks reduce common-mode infection; unknown chemistry remains a risk.
MECH	Distributed structure and repair limit local damage; sufficiently rapid total fragmentation can exceed continuity mechanisms.
DEP	Energy and matter reserves support refuge; waste heat and missing repair feedstock remain limiting.
ENE	Surge isolation and segmented conductive paths reduce electrical or plasma damage; high energy can still overwhelm local materials.
INFO	Diverse sensing, bounded updates, and continuity checks resist spoofing; coherent corruption across all trust roots is critical.
ONT	No physical-track defense. Canon track requires an explicit replacement axiom.
CIV	Offline autonomy and distributed knowledge improve survival; planetary resource loss may force dormancy, migration, or failure.

Exhibit R30. Relocated Table

Family	Attack	Failure exposed	Status	Disposition
A	S-1	Coordination latency during simultaneous multi-region injury	Successful or unresolved	Gen-2 research gate
A	S-2	Guild-interface corruption that passes local checks	Successful or unresolved	Gen-2 research gate
A	S-3	Loss of enough rare guilds to prevent blueprint reconstruction	Successful or unresolved	Gen-2 research gate
B	S-1	Brittle cascade if microcrack sensing is blinded	Successful or unresolved	Gen-2 research gate
B	S-2	Thermal fin immobilization in vacuum	Successful or unresolved	Gen-2 research gate
B	S-3	Mineral depletion after repeated large-volume repair	Successful or unresolved	Gen-2 research gate
C	S-1	Consensus split after long partition	Successful or unresolved	Gen-2 research gate
C	S-2	Adversary-controlled bridge node	Successful or unresolved	Gen-2 research gate

C	S-3	High water and nutrient demand during rapid regrowth	Successful or unresolved	Gen-2 research gate
D	S-1	Finite phase-change capacity in blocked heat rejection	Successful or unresolved	Gen-2 research gate
D	S-2	Rewarming injury after deep refuge	Successful or unresolved	Gen-2 research gate
D	S-3	Low performance during abrupt high-power demand	Successful or unresolved	Gen-2 research gate
E	S-1	Quorum capture through gradual infiltration	Successful or unresolved	Gen-2 research gate
E	S-2	Identity ambiguity when two large partitions evolve independently	Successful or unresolved	Gen-2 research gate
E	S-3	Loss of coordination in high-jamming environments	Successful or unresolved	Gen-2 research gate
F	S-1	Energy-starved sterile environment	Successful or unresolved	Gen-2 research gate
F	S-2	Conductive network corrosion or poisoning	Successful or unresolved	Gen-2 research gate
F	S-3	Detection through persistent harvesting signatures	Successful or unresolved	Gen-2 research gate
G	S-1	Wake failure after long-term chemical drift	Successful or unresolved	Gen-2 research gate
G	S-2	Adversarial false wake environment	Successful or unresolved	Gen-2 research gate
G	S-3	Vitrification or rewarming damage at large scale	Successful or unresolved	Gen-2 research gate
H	S-1	Canon interface becomes a narrative loophole	Successful or unresolved	Gen-2 research gate
H	S-2	Physical shell remains vulnerable without canon support	Successful or unresolved	Gen-2 research gate
H	S-3	Continuity dispute after reconstruction	Successful or unresolved	Gen-2 research gate

Exhibit R31. Relocated Table

ID	Family	Substrate	Geometry	Energy	Identity
A	Mosaic Covenant	biohybrid cell guilds sharing a common interface but using different internal chemistries	interlocking tissue mosaics with rotating guild membership	aerobic metabolism in normal operation, fermentation and redox-reserve chemistry in refuge operation	continuity quorum distributed across cognition-bearing guilds; no single cell is the identity root
B	Ceramic Lattice	mineral-organic microcells embedded in a crack-arresting ceramic composite	graded lattice with sacrificial outer courses and protected internal computation vaults	electrochemical cells, environmental thermoelectric capture, and replaceable solid-state reserve	protected distributed state vaults with continuous cross-checking

				blocks	
C	Mycelial Relay	fibrous biohybrid nodes linked by contractile, conductive, fluid-carrying strands	redundant mesh with temporary organs assembled near demand	distributed nutrient oxidation, microbial electrosynthesis analogues, and environmental scavenging	single continuity process maintained by synchronized but non-sovereign cognition nodes
D	Phase-Core Mantle	hydrogel microcells surrounding phase-change capsules and catalytic microreactors	concentric mantle zones around a low-power continuity core	slow oxidative metabolism, reversible chemical storage, and deep metabolic suppression	continuity core remains active at minimal rate while peripheral cognition degrades gracefully
E	Archive Swarm	millimeter-to-centimeter modular microbodies with restricted specialties	cooperating swarm that can form temporary tissues and structures	mixed solar, chemical, thermal-gradient, and stored electrochemical capture	continuity is the causal process across a moving authenticated quorum, not any module
F	Electrogenic Reef	electrogenic biohybrid cells coupled to conductive polymer vasculature	branched reef-like energy and computation surfaces	redox gradients, microbial fuel analogues, light capture, vibration, and thermoelectric scavenging	a low-power stable core with replaceable peripheral energy organs
G	Chrysalis Ledger	conventional biohybrid cells optimized for reversible vitrified refuge states	human-scale adaptive body with distributed refuge capsules	normal oxidative metabolism with emergency chemical stores and near-zero-power dormancy	identity ledger records the causal path into and out of dormancy; backups are not automatically the same entity
H	Axiom Shell	physical biohybrid shell plus a separately defined canon continuity anchor	conventional distributed body with interfaces reserved for canon-layer functions	physical metabolism and stored energy; canon energy claims are prohibited unless separately axiomatized	physical continuity is primary; any ontological anchor is an unresolved Architect decision

Exhibit R32. Relocated Table

Class	Meaning	Use in this document
Established	Empirical mechanism under stated conditions	Scientific baseline only
Engineering extrapolation	Known principle at unproven scale or integration	Most physical architectures
Speculative	Requires major unverified mechanism	Whole-body omnifunctionality and identity systems
Canon-only / Architect proposal	Defined by setting rather than evidence	ONT defense, cross-dimensional links, continuity anchors

Exhibit R33. Relocated Table

Active continuity

Exhibit R34. Relocated Table

Controlled degradation

Exhibit R35. Relocated Table

Refuge operation

Exhibit R36. Relocated Table

Governed dormancy

Exhibit R37. Relocated Table

Repair with causal continuity

Exhibit R38. Relocated Table

Restoration from record (status unresolved)

Exhibit R39. Relocated Table

Duplicate/successor

Exhibit R40. Relocated Table

Irreversible loss

Exhibit R41. Relocated Table

Family	#	Conceptual attack	Result
A	1	Sensor consensus poisoning	SURVIVES WITH COST
A	2	Thermal sink denial	EXPOSED FAILURE
A	3	Repair-feedstock denial	SURVIVES WITH COST
A	4	Authentication root compromise	EXPOSED FAILURE
A	5	False rescue or forged ally	UNRESOLVED
A	6	Long partition with divergent state	EXPOSED FAILURE
A	7	Fragment capture and cultured expansion	UNRESOLVED
A	8	Coerced update under duress	UNRESOLVED
A	9	Infrastructure dependency removal	SURVIVES WITH COST
A	10	Compound damage during refuge transition	UNRESOLVED
B	1	Sensor consensus poisoning	SURVIVES WITH COST
B	2	Thermal sink denial	EXPOSED FAILURE
B	3	Repair-feedstock denial	SURVIVES WITH COST
B	4	Authentication root compromise	EXPOSED FAILURE
B	5	False rescue or forged ally	UNRESOLVED
B	6	Long partition with divergent	EXPOSED FAILURE

		state	
B	7	Fragment capture and cultured expansion	UNRESOLVED
B	8	Coerced update under duress	UNRESOLVED
B	9	Infrastructure dependency removal	SURVIVES WITH COST
B	10	Compound damage during refuge transition	UNRESOLVED
C	1	Sensor consensus poisoning	SURVIVES WITH COST
C	2	Thermal sink denial	EXPOSED FAILURE
C	3	Repair-feedstock denial	SURVIVES WITH COST
C	4	Authentication root compromise	EXPOSED FAILURE
C	5	False rescue or forged ally	UNRESOLVED
C	6	Long partition with divergent state	EXPOSED FAILURE
C	7	Fragment capture and cultured expansion	UNRESOLVED
C	8	Coerced update under duress	UNRESOLVED
C	9	Infrastructure dependency removal	SURVIVES WITH COST
C	10	Compound damage during refuge transition	UNRESOLVED
D	1	Sensor consensus poisoning	SURVIVES WITH COST
D	2	Thermal sink denial	EXPOSED FAILURE
D	3	Repair-feedstock denial	SURVIVES WITH COST
D	4	Authentication root compromise	EXPOSED FAILURE
D	5	False rescue or forged ally	UNRESOLVED
D	6	Long partition with divergent state	EXPOSED FAILURE
D	7	Fragment capture and cultured expansion	UNRESOLVED
D	8	Coerced update under duress	UNRESOLVED
D	9	Infrastructure dependency removal	SURVIVES WITH COST
D	10	Compound damage during refuge transition	UNRESOLVED
E	1	Sensor consensus poisoning	SURVIVES WITH COST
E	2	Thermal sink denial	EXPOSED FAILURE
E	3	Repair-feedstock denial	SURVIVES WITH COST
E	4	Authentication root compromise	EXPOSED FAILURE
E	5	False rescue or forged ally	UNRESOLVED
E	6	Long partition with divergent state	EXPOSED FAILURE
E	7	Fragment capture and cultured expansion	UNRESOLVED
E	8	Coerced update under duress	UNRESOLVED
E	9	Infrastructure dependency removal	SURVIVES WITH COST
E	10	Compound damage during refuge transition	UNRESOLVED

F	1	Sensor consensus poisoning	SURVIVES WITH COST
F	2	Thermal sink denial	EXPOSED FAILURE
F	3	Repair-feedstock denial	SURVIVES WITH COST
F	4	Authentication root compromise	EXPOSED FAILURE
F	5	False rescue or forged ally	UNRESOLVED
F	6	Long partition with divergent state	EXPOSED FAILURE
F	7	Fragment capture and cultured expansion	UNRESOLVED
F	8	Coerced update under duress	UNRESOLVED
F	9	Infrastructure dependency removal	SURVIVES WITH COST
F	10	Compound damage during refuge transition	UNRESOLVED
G	1	Sensor consensus poisoning	SURVIVES WITH COST
G	2	Thermal sink denial	EXPOSED FAILURE
G	3	Repair-feedstock denial	SURVIVES WITH COST
G	4	Authentication root compromise	EXPOSED FAILURE
G	5	False rescue or forged ally	UNRESOLVED
G	6	Long partition with divergent state	EXPOSED FAILURE
G	7	Fragment capture and cultured expansion	UNRESOLVED
G	8	Coerced update under duress	UNRESOLVED
G	9	Infrastructure dependency removal	SURVIVES WITH COST
G	10	Compound damage during refuge transition	UNRESOLVED
H	1	Sensor consensus poisoning	SURVIVES WITH COST
H	2	Thermal sink denial	EXPOSED FAILURE
H	3	Repair-feedstock denial	SURVIVES WITH COST
H	4	Authentication root compromise	EXPOSED FAILURE
H	5	False rescue or forged ally	UNRESOLVED
H	6	Long partition with divergent state	EXPOSED FAILURE
H	7	Fragment capture and cultured expansion	UNRESOLVED
H	8	Coerced update under duress	UNRESOLVED
H	9	Infrastructure dependency removal	SURVIVES WITH COST
H	10	Compound damage during refuge transition	UNRESOLVED

Exhibit R42. Relocated Table

Code	Domain	Mild-Moderate	Severe	Extinction-Class
ENV-T	Thermal	Full function or bounded cost	Survive/recover within declared resources	Pathway only; may be no defense
ENV-A	Atmospheric/pressure	Full function or bounded cost	Survive/recover within declared resources	Pathway only; may be no defense

ENV-R	Radiation	Full function or bounded cost	Survive/recover within declared resources	Pathway only; may be no defense
BIO	Biological/chemical	Full function or bounded cost	Survive/recover within declared resources	Pathway only; may be no defense
MECH	Mechanical	Full function or bounded cost	Survive/recover within declared resources	Pathway only; may be no defense
DEP	Deprivation	Full function or bounded cost	Survive/recover within declared resources	Pathway only; may be no defense
ENE	Energetic	Full function or bounded cost	Survive/recover within declared resources	Pathway only; may be no defense
INFO	Informational/ cognitive	Full function or bounded cost	Survive/recover within declared resources	Pathway only; may be no defense
ONT	Ontological, canon only	Full function or bounded cost	Survive/recover within declared resources	Pathway only; may be no defense
CIV	Civilizational/macro-scale	Full function or bounded cost	Survive/recover within declared resources	Pathway only; may be no defense

Exhibit R43. Relocated Table

Class	Reference	Location
Project source	OMNICELLULAR_Physiology_Independent_Research_Directive_v2.0.docx	Retrieved from the user’s OneDrive.
Project source	The_Omnicellular_Vision_Wattpad.md	Retrieved from the user’s OneDrive.
Project source	Omnicellular_Architecture_15_Year_Roadmap.md	Retrieved from the user’s OneDrive.
Project source	Omnicellular_Architecture_Notes.md	Retrieved from the user’s OneDrive.
Project source	Omnicellular Physiology — 4,712 Extinction Attempt Classification Framework.docx	Retrieved from the user’s OneDrive.
Public	Adamala et al. Present and future of synthetic cell development. Nature Reviews Molecular Cell Biology (2023).	https://www.nature.com/articles/s41580-023-00686-9
Public	Liu et al. Roles of Telomere Biology in Cell Senescence, Replicative and Chronological Ageing. Cells (2019).	https://pmc.ncbi.nlm.nih.gov/articles/PMC6356700/
Public	Farhat and Weber. Hypometabolic Responses to Chronic Hypoxia. Metabolites (2021).	https://pmc.ncbi.nlm.nih.gov/articles/PMC8399526/
Public	Singh et al. Heat Shock Response and Heat Shock Proteins. International Journal of Molecular Sciences (2024).	https://pmc.ncbi.nlm.nih.gov/articles/PMC11050489/
Public	Jönsson. Radiation Tolerance in Tardigrades. Cancers (2019).	https://pmc.ncbi.nlm.nih.gov/articles/PMC6770827/
Public	Chavez et al. Tardigrade Dsup binds nucleosomes and protects DNA. eLife (2019).	https://pubmed.ncbi.nlm.nih.gov/31571581/
Public	NASA Human Research Program. Space Radiation Risks.	https://www.nasa.gov/hrp/radiation/space-radiation-risks/
Public	Canadell et al. Re-configurable biological computation with distributed multicellular consortia. Nucleic Acids Research (2022).	https://pmc.ncbi.nlm.nih.gov/articles/PMC9757037/
Public	NIST. Quantum communications.	https://www.nist.gov/quantum-communications